

LAPORAN JOBSHEET
PEMROGRAMAN BERBASIS FRAMEWORK



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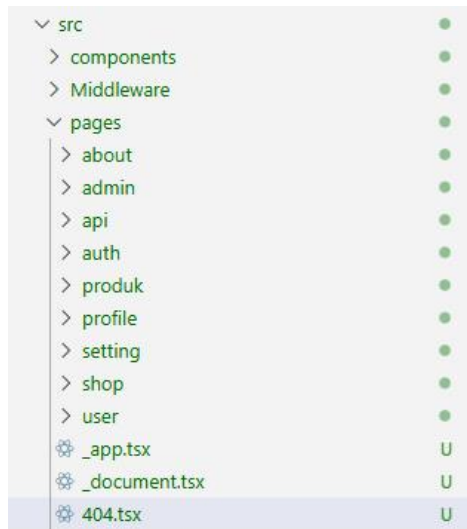
Kelas: TI 3E

POLITEKNIK NEGERI MALANG TAHUN 2026

2A. Optimasi Gambar Lokal (Public Folder)

- Studi Kasus:
- Mengganti tag pada halaman 404 dengan next/image. Langkah:

- Buka file src/pages/404.tsx



- Modifikasi line 7 menjadi line 8-11

```
17.praktikum-Optimize > my-app > src > pages > 404.tsx > ...
1  import Image from "next/image";
2  import styles from "@styles/404.module.scss";
3
4  Windsurf: Refactor | Explain | Generate JSDoc | X
5  const Custom404 = () => {
6    return (
7      <div className={styles.error}>
8        { /*  */ }
9        <Image
10          src="/page-not-found.png"
11          alt="404"
12          width={400}
13          height={200}
14          className={styles.error__image}
15        />
16        <h1>404 - Halaman Tidak Ditemukan</h1>
17        <p>Maaf, halaman yang Anda cari tidak ada.</p>
18      </div>
19    );
20  };
21  export default Custom404;
```

```

c > pages > 404.tsx > ...
1  import Image from "next/image";
2  import styles from "@styles/404.module.scss";
3
Windsurf: Refactor | Explain | Generate JSDoc | X
4  const Custom404 = () => {
5      return (
6          <div className={styles.error}>
7              { /*  */ }
8              <Image
9                  src="/page-not-found.png"
10                 alt="404"
11                 width={400}
12                 height={200}
13                 className={styles.error__image}
14             />
15             <h1>404 - Halaman Tidak Ditemukan</h1>
16             <p>Maaf, halaman yang Anda cari tidak ada.</p>
17         </div>
18     );
19 };
20
21 export default Custom404;
22

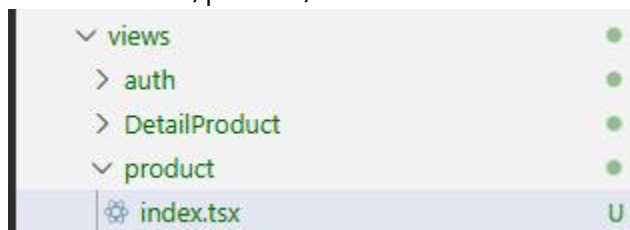
```

🚀 Hasil:

- Warning hilang
- Image dioptimasi otomatis
- Mengurangi bandwidth
- Mendukung lazy loading otomatis

🔧 B. Optimasi Gambar Remote (External URL)

- Buka file views/product/index.tsx



- Modifikasi file index.tsx

```

17.praktikum-Optimize > my-app > src > views > product > index.tsx > TampilanProduk > products.map() callback
1  import styles from "../../pages/produk/product.module.scss";
2  import Link from "next/link";
3  import Image from "next/image";
4

```

```

17.praktikum-Optimize > my-app > src > views > produk > index.tsx > TampilanProduk > products.map() callback
13  const TampilanProduk = ({ products }: { products: ProductType[] }) => {
17    <div className={styles.produk_content}>
18      {products.length > 0 ? (
19        <>
20          {products.map((products: ProductType) => (
21            <Link href={` /produk/${products.id}`} key={products.id} className={styles.produk_content_item}>
22              <div className={styles.produk_content_item_image}>
23                <Image src={products.image} alt={products.name} width={200} height={200} />
24              </div>
25              <h4 className={styles.produk_content_item_name}>
26                {products.name}
27              </h4>
28              <p className={styles.produk_content_item_category}>
29                {products.category}
30              </p>
31              <p className={styles.produk_content_item_price}>
32                Rp {products.price.toLocaleString("id-ID")}
33              </p>
34            </Link>
35          )]}
36        </>
37      ) : (

```

Note : dikarenakan gambar diambil dari url tertentu maka konfigurasi berbeda

```

src > pages > views > produk > index.tsx > ...
1  import styles from "../../produk/product.module.scss";
2  import Link from "next/link";
3  import Image from "next/image";
4

```

```

2  Windsurf: Refactor | Explain | Generate JSDoc | X
3  const TampilanProduk = ({ products }: { products?: ProductType[] }) => {
4    return (
5      <div className={styles.produk}>
6        <h1 className={styles.produk_title}>Daftar Produk</h1>
7        <div className={styles.produk_content}>
8          {products && products.length > 0 ? (
9            <>
10              {products.map((products: ProductType) => [
11                <Link href={` /produk/${products.id}`} key={products.id} classNam
12                  <div className={styles.produk_content_item_image}>
13                    <img src={products.image} alt={products.name} width={200} />
14                  </div>
15                <h4 className={styles.produk_content_item_name}>
16                  {products.name}
17                </h4>
18                <p className={styles.produk_content_item_category}>
19                  {products.category}
20                </p>
21                <p className={styles.produk_content_item_price}>
22                  Rp {products.price.toLocaleString("id-ID")}
23                </p>
24              ])}
25            </>
26          ) : (

```

- Buka file next.config.js

```

17.praktikum-Optimize > my-app > JS next.config.js > nextConfig > images > remotePatterns > hostname
1  /** @type {import('next').NextConfig} */
2  const nextConfig = {
3    reactStrictMode: true,
4    images: {
5      remotePatterns: [
6        {
7          protocol: "https",
8          hostname: "assets.adidas.com",
9          port: "",
10         pathname: "**",
11       },
12     ],
13   },
14 }
15
16
17 module.exports = nextConfig
18

```

```

23  /** @type {import('next').NextConfig} */
24  const nextConfig = {
25    reactStrictMode: true,
26    images: {
27      remotePatterns: [
28        {
29          protocol: "https",
30          hostname: "assets.adidas.com",
31          port: "",
32          pathname: "**",
33        },
34      ],
35    },
36  }
37
38  module.exports = nextConfig

```

Hasil:

- Gambar di-proxy melalui `/_next/image`
- Performa lebih optimal
- Kompresi otomatis

✈ PRAKTIKUM 2 – Font Optimization

🚩 A. Menggunakan next/font

- Buka file index.tsx pada folder Appshell/index.tsx dan modifikasi

```
17.praktikum-Optimize > my-app > src > components > layouts > Appshell > index.tsx > ...
1  import { useRouter } from "next/router";
2  import Navbar from "../Navbar";
3  import { Roboto } from "next/font/google";
4
5  const disableNavbar = ['/auth/login', '/auth/register', '/404'];
6
7  type AppShellProps = {
8    children: React.ReactNode;
9  }
10
11  const roboto = Roboto({
12    subsets: ["latin"],
13    weight: ["400", "500", "700"],
14  });
15
16  const AppShell = (props: AppShellProps) => {
17
18    const { children } = props;
19    const { pathname } = useRouter();
20
21    return (
22      <main className={roboto.className}>
23        {!disableNavbar.includes(pathname) & <Navbar />}
24        {children}
25      </main>
26    );
27  };
28
29  export default AppShell;
```



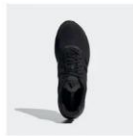
```

1  import Navbar from "../navbar";
2  import {useRouter} from "next/router";
3  import { Roboto } from "next/font/google";
4
5  const disableNavbar = ['/auth/login', '/auth/register'];
6
7  type AppShellProps = {
8    children: React.ReactNode;
9  };
10
11 const roboto = Roboto({
12   subsets: ["latin"],
13   weight: ["400", "500", "700"],
14 });
15
16 const AppShell = (props: AppShellProps) => {
17   const { children } = props;
18   const { pathname } = useRouter();
19   return (
20     <main className={roboto.className}>
21       <!disableNavbar.includes(pathname) && <Navbar />
22       {children}
23     </main>
24   );
25 }
26
27 export default AppShell;

```

- Jalankan browser localhost:3000/produk maka font akan berubah menjadi roboto untuk mengecek fontnya bisa menggunakan extension FontFinder

Daftar Produk



Sepatu Duramo SL

Meris Shoes

Rp 9.000.000



Sepatu Duramo SL

Meris Shoes

Rp 900.000,00



SEPATU SAMBA OG

Meris Shoes

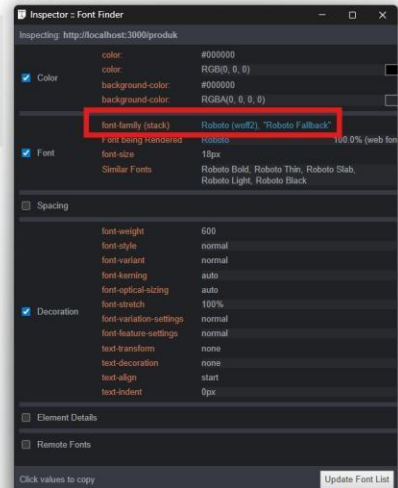
Rp 2.000.000,00

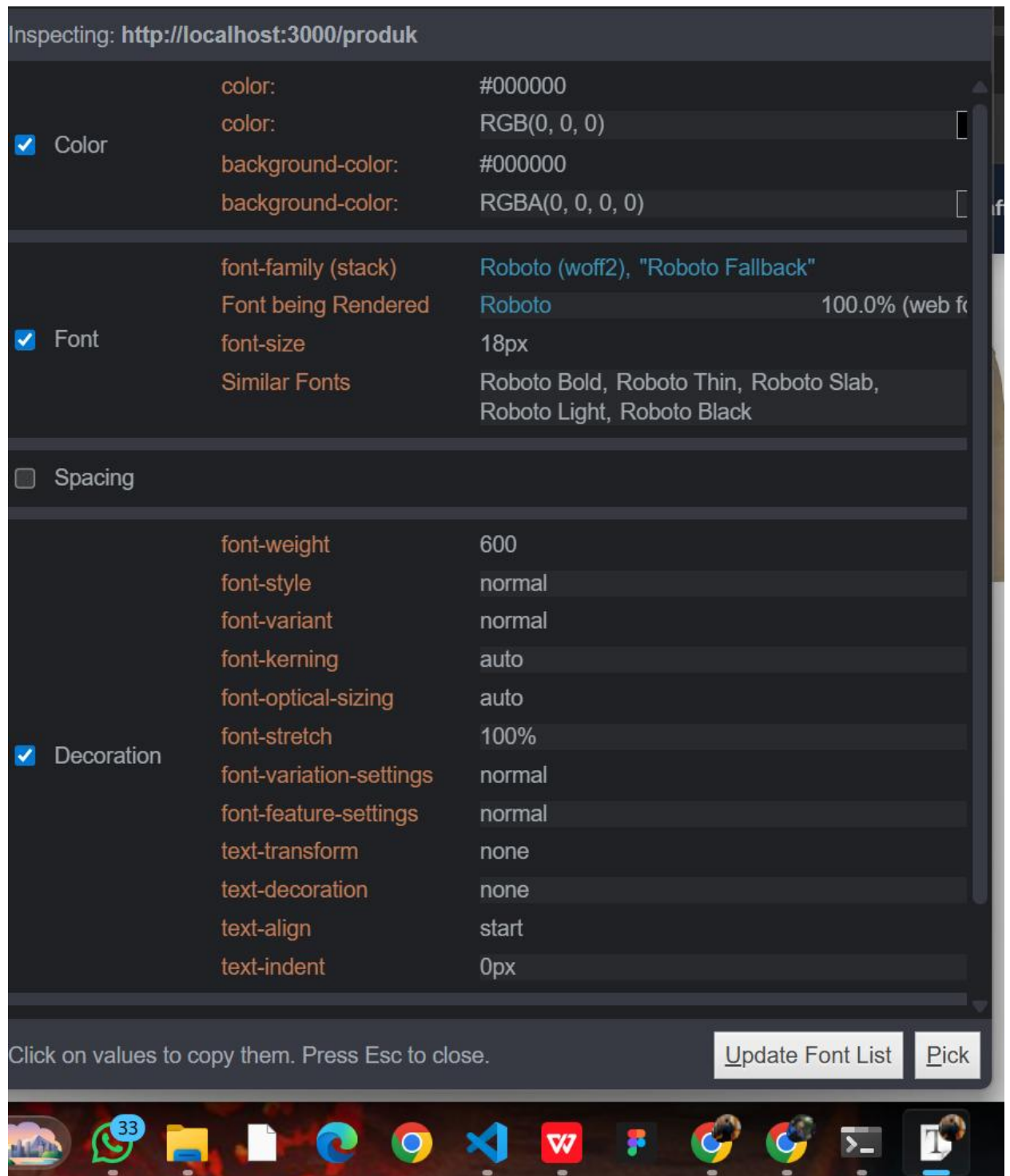


sepatu handball spesial

Meris Shoes

Rp 1.700.000





Hasil:

- Tidak perlu load dari CDN manual
- Tidak blocking render
- Performance meningkat
- Tidak terjadi FOUT (Flash of Unstyled Text)

🔴 B. Menggunakan next/script

- Buka file index.tsx pada folder layouts/Navbar dan modifikasi

```
17 praktikum-Optimize > my-app > src > components > layouts > navbar > index.tsx > Navbar
1 import Script from 'next/dist/client/script';
2 import styles from './navbar.module.css';
3 import { signIn, signOut, useSession } from "next-auth/react"
4
Windsurf: Refactor | Explain | Generate JSDoc | X
5 const Navbar = () => {
6   const {data}:any = useSession()
7   //const { data: session } = useSession()
8   // console.log("session", session)
9   return (
10    <div className={styles.navbar}>
11      { /* <div className={styles.navbar__brand}>
12        MyApp
13      </div> */}
14
15      <div className={styles.navbar__brand} id="title"></div>
16      <Script id="title-script" strategy='lazyOnload' >
17        {`document.getElementById('title').innerHTML = 'MyApp';`}
18      </Script>
```

```

src > pages > components > layouts > navbar > index.tsx > Navbar
1  import Script from "next/script";
2  import styles from "./navbar.module.css";
3  import { signIn, signOut, useSession } from "next-auth/react";
4  import Image from "next/image";
   Windsurf: Refactor | Explain | Generate JSDoc | X
5  const Navbar = () => {
6      const { data }: any = useSession();
7      //const { data: session } = useSession()
8      // console.log("session", session)
9      return (
10         <div className={styles.navbar}>
11             <div className={styles.navbar__brand} id="title"></div>
12             <Script id="title-script" strategy="lazyOnload">
13                 | {`document.getElementById('title').innerHTML = 'My App';`}
14             </Script>
15
16             <div className={styles.navbar__right}>
17                 {data ? (
18                     <>
19                         <div className={styles.navbar__user}>
20                             Welcome, {data.user?.fullname}
21                             {data.user?.image && (
22                                 <Image
23                                     src={data.user.image}
24                                     alt={data.user.fullname}
25                                     className={styles.navbar__user_image}
26                                     width={42}
27                                     height={42}
28                                 />

```

- Pada kasus diatas kita merubah line 11 menggunakan model Typescript dapat terlihat ketika kita refresh web produk tulisan myApp akan terlihat berkedip
- Perbedaan mendasar antara Line 11-13 dan Line 15-18 pada file index.tsx Anda terletak pada metode rendering teks dan interaksi dengan DOM (Document Object Model). Berikut adalah rincian perbedaannya:

I. Metode Rendering

- a. Line 11-13 (Standard React/JSX): Ini adalah cara standar React. Teks "MyApp" ditulis langsung di dalam tag div. React akan langsung merender teks ini ke dalam HTML saat komponen dimuat.
- b. Line 15-18 (Next.js Script & Manual DOM): Menggunakan komponen <Script> dari Next.js dengan strategi lazyOnload. Teks tidak ada di dalam HTML saat awal dimuat, melainkan baru "disuntikkan" secara manual menggunakan perintah JavaScript `document.getElementById('title').innerHTML = 'MyApp';` setelah script tersebut diunduh di latar belakang.

II. Performa dan SEO

- a. Line 11-13: Sangat baik untuk SEO karena teks "MyApp" langsung terbaca oleh robot pencari (crawler) dalam kode sumber HTML.
- b. Line 15-18: Kurang baik untuk SEO untuk konten penting karena teks baru muncul setelah JavaScript dijalankan. Strategi lazyOnload berarti script ini dijalankan paling akhir setelah semua sumber daya utama selesai dimuat, sehingga mungkin ada jeda waktu (delay) sebelum teks muncul di layar.

III. Keamanan (XSS)

- a. Line 11-13: Aman karena React secara otomatis melakukan escape pada string untuk mencegah serangan Cross-Site Scripting (XSS).
- b. Line 15-18: Memiliki risiko keamanan lebih tinggi karena menggunakan `.innerHTML`. Jika data yang dimasukkan berasal dari input user, ini bisa dimanfaatkan untuk menyuntikkan skrip berbahaya.

z C. Strategi Script

Strategy	Fungsi
beforeInteractive	Sebelum halaman interaktif
afterInteractive	Setelah halaman interaktif
lazyOnload	Setelah semua selesai
worker	Web worker

🚩 Hasil:

- Script tidak blocking
 - Cocok untuk Google Analytics
 - Performa lebih ringan
-

🔗 PRAKTIKUM 4 – Optimasi Avatar dengan next/image

- Buka file index.tsx pada folder layouts/navbar dan modifikasi :

17.praktikum-Optimize > my-app > src > components > layouts > navbar > index.tsx > [🔍] Navbar

```
1 import Script from 'next/dist/client/script';
2 import Image from 'next/image';
3 import styles from './navbar.module.css';
4 import { signIn, signOut, useSession } from "next-auth/react"
5
6 Windsurf: Refactor | Explain | Generate JSDoc | X
7 const Navbar = () => {
8   const {data}:any = useSession()
9   //const { data: session } = useSession()
10  // console.log("session", session)
11  return (
12    <div className={styles.navbar}>
13      <div className={styles.navbar__brand}>
14        MyApp
15      </div>
16      <div className={styles.navbar__brand} id="title"></div>
17      <Script id="title-script" strategy='lazyOnload' >
18        {`document.getElementById('title').innerHTML = 'MyApp';`}
19      </Script>
20
21      <div className={styles.navbar__right}>
22        {data ? (
23          <>
24            <div className={styles.navbar__user}>
25              Welcome, {data.user?.fullname}
26              {data?.user?.image && (
27                // <img
28                <Image
29                  width={50} height={50}
30                  src={data.user.image}
31                  alt={data.user.fullname}
32                  className={styles.navbar__user_image}
33                />
34              )}
35            </div>
```

```

1 import Script from "next/script";
2 import styles from "./navbar.module.css";
3 import { signIn, signOut, useSession } from "next-auth/react";
4 import Image from "next/image";
5
6 const Navbar = () => {
7   const { data } = useSession();
8   //const { data: session } = useSession()
9   // console.log("session", session)
10
11   return (
12     <div className={styles.navbar}>
13       <div className={styles.navbar_brand} id="title"></div>
14       <Script id="title-script" strategy="lazyOnload">
15         {`document.getElementById('title').innerHTML = 'My App';`}
16       </Script>
17
18       <div className={styles.navbar__right}>
19         {data ? (
20           <>
21             <div className={styles.navbar_user}>
22               Welcome, {data.user?.fullname}
23               {data.user?.image && (
24                 <Image
25                   src={data.user.image}
26                   alt={data.user.fullname}
27                   className={styles.navbar_user_image}
28                   width={50}
29                   height={50}
30                 />
31               )}
32             </div>
33           </>
34         ) : null}
35       </div>
36     </div>
37   );
38 }

```

- Tambahkan hostname Google:

```

17.praktikum-Optimize > my-app > JS next.config.js > ...
1  /** @type {import('next').NextConfig} */
2  const nextConfig = {
3    reactStrictMode: true,
4    images: {
5      remotePatterns: [
6        {
7          protocol: "https",
8          hostname: "assets.adidas.com",
9          port: "",
10         pathname: "**",
11       },
12       {
13         protocol: "https",
14         hostname: "lh3.googleusercontent.com",
15         pathname: "**",
16       },
17     ],
18   },
19 };
20
21 module.exports = nextConfig;
22

```



```

JS next.config.js > nextConfig > images
1  /** @type {import('next').NextConfig} */
2  const nextConfig = {
3    reactStrictMode: true,
4
5    images: {
6      remotePatterns: [
7        {
8          protocol: "https",
9          hostname: "assets.adidas.com",
10         pathname: "**",
11       },
12       {
13         protocol: "https",
14         hostname: "lh3.googleusercontent.com",
15         pathname: "**",
16       },
17     ],
18   },
19
20   turbopack: {}, // TAMBAHKAN INI
21 };
22
23 module.exports = nextConfig;

```

Tugas Praktikum

- Optimasi semua image di project menggunakan next/image

```

1  import Script from "next/dist/client/script";
2  import styles from "../navbar.module.css";
3  import { signIn, signOut, useSession } from "next-auth/react";
4  import Image from "next/image";
5
6  const Navbar = () => {
7    const { data: session } = useSession();
8    //const { data: session } = useSession();
9    console.log("session", data); // { session: { user: { name: 'Van Daffa Putra Liandhie', email: 'yandaffa19@gmail.com', image: null } } }
10   return (
11     <div className={styles.navbar}>
12       <div className={styles.navbar_brand}>
13         My App
14       </div>
15       <div className={styles.navbar_brand} id="title"></div>
16       <Script id="title-script" strategy="lazyOnload">
17         {document.getElementById("title").innerHTML = "My App";}
18       </Script>
19       <div className={styles.navbar_right}>
20         {data ? (
21           <div className={styles.navbar_user}>
22             Welcome, {data.user.fullName}
23             <div data-user-image={data.user.image} && {
24               <Image
25                 src={data.user.image}
26                 alt={data.user.fullName}
27                 width={50} height={50}
28                 className={styles.navbar_user_image}
29               />
30             )}
31           </div>
32           <button
33             className={styles.navbar_button + " " + styles["navbar_button--danger"]}
34             onClick={() => signOut()}
35           >
36             Sign Out
37           </button>
38         ) : (
39           <button
40             className={styles.navbar_button + " " + styles["navbar_button--primary"]}
41             onClick={() => signIn()}
42           >
43             Sign In
44           </button>
45         )}
46       </div>
47     </div>
48   );
49 }
50
51 export default Navbar;

```



```

1 import Image from 'next/image';
2 import Link from 'next/link';
3 import styles from "../../pages/product/product.module.scss";
4 type ProductType = {
5   id: string;
6   name: string;
7   price: number;
8   images: string; // Sesuai dengan Firebase
9   category: string;
10 };
11
12 const TampilanProduk = ({products}: {products: ProductType[]}) => {
13   return (
14     <div className={styles.produk}>
15       <h1 className={styles.produk_title}>Daftar Produk</h1>
16       <div className={styles.produk_content}>
17         {products.length === 0 ? (
18           <div className={styles.produk_content_skeleton}>
19             <div className={styles.produk_content_skeleton_image}></div>
20             <div className={styles.produk_content_skeleton_name}></div>
21             <div className={styles.produk_content_skeleton_category}></div>
22             <div className={styles.produk_content_skeleton_price}></div>
23           </div>
24         ) : (
25           <div>
26             {products.map((product: ProductType, index: number) => (
27               <Link href={`/${product.id}`} key={product.id} className={styles.produk_content_item}>
28                 /*<div key={product.id} className={styles.produk_content_item}>*/
29                 <div className={styles.produk_content_item_image}>
30                   <Image src={product.images} alt={product.name} width={200} height={200} priority={index < 4} />
31                 </div>
32                 <h4 className={styles.produk_content_item_name}>
33                   {product.name}
34                 </h4>
35                 <p className={styles.produk_content_item_category}>
36                   {product.category}
37                 </p>
38                 <p className={styles.produk_content_item_price}>
39                   Rp {product.price.toLocaleString('id-ID')}
40                 </p>
41               </Link>
42             ))}
43           </div>
44         )}
45       </div>
46     </div>
47   );
48 };
49
50 export default TampilanProduk;
// export default

```

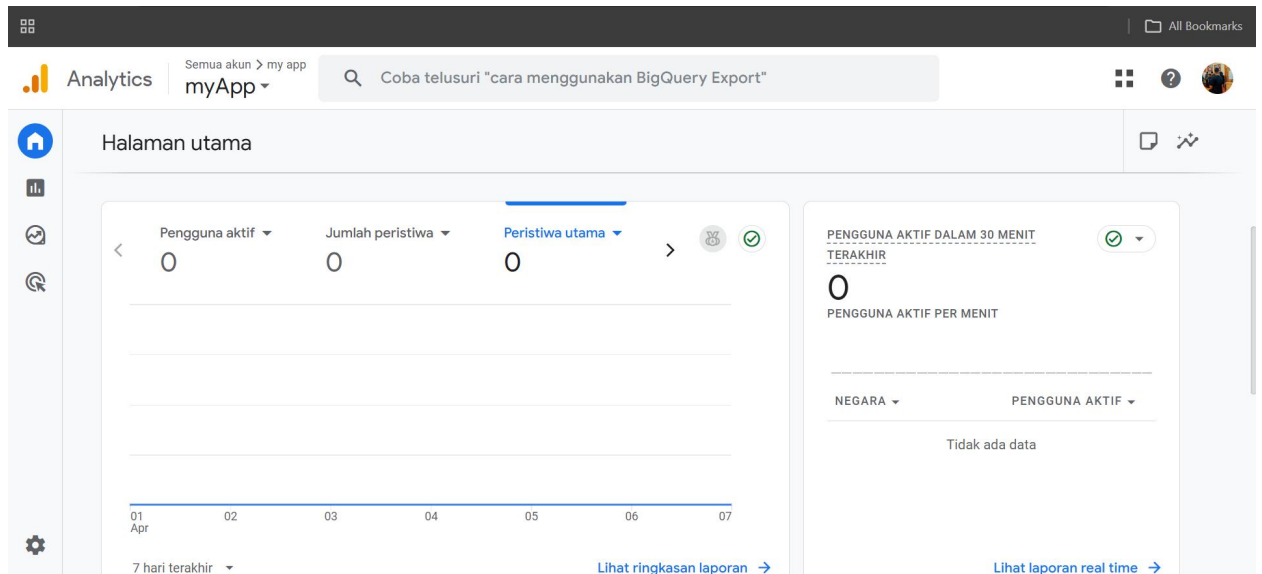
2. Gunakan minimal 1 font dari next/font

```

11 const roboto = Roboto({
12   subsets: ["latin"],
13   weight: ["400", "500", "700"],
14 });
15

```

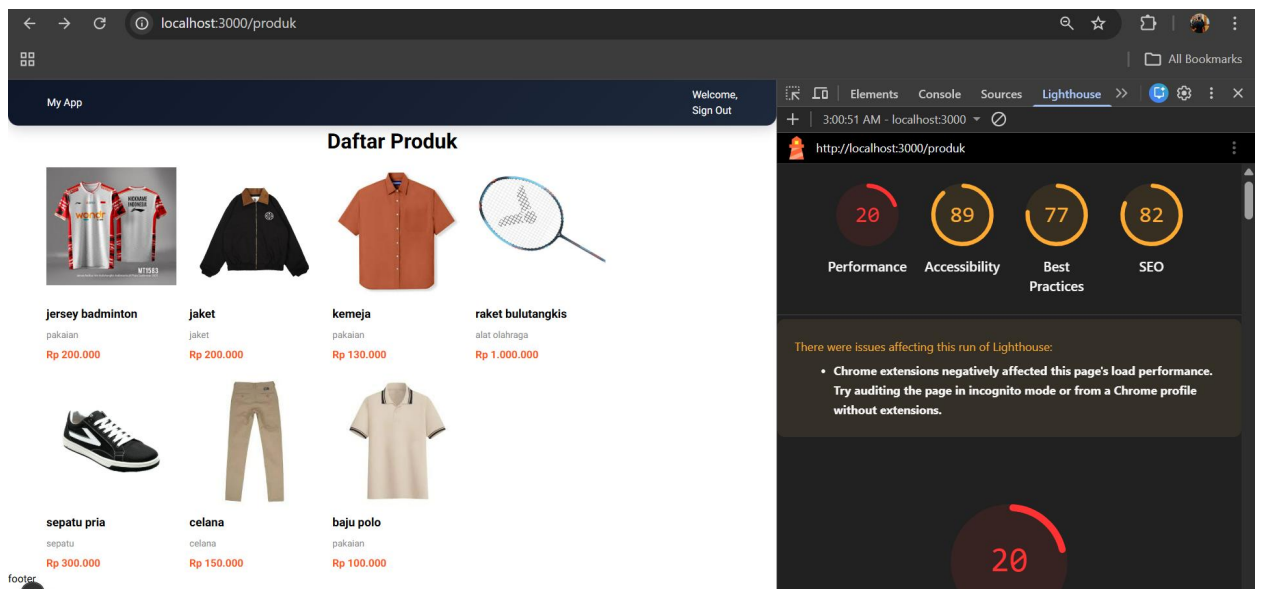
3. Tambahkan script Google Analytics menggunakan next/script

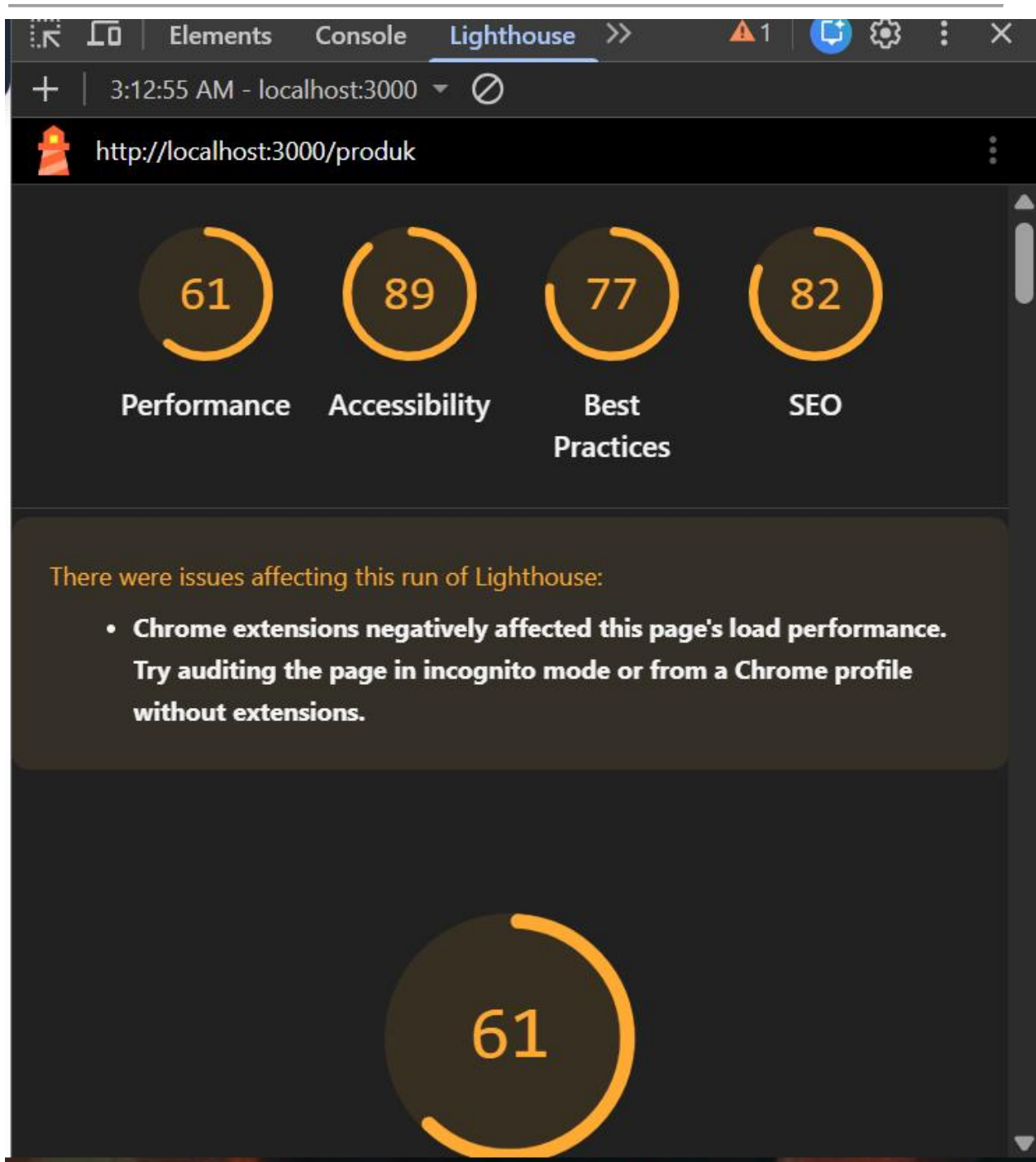


4. Terapkan dynamic import pada minimal 1 komponen

```
9 const DetailProduk = dynamic(() => import("../views/DetailProduct"), {
10   loading: () => <p>Loading detail produk...</p>,
11 });
```

5. Dokumentasikan perubahan performa (screenshot Lighthouse)





Refleksi & Diskusi

1. Mengapa `<img>` biasa tidak optimal?

Karena tidak memiliki fitur bawaan seperti kompresi otomatis, lazy loading cerdas, dan penyesuaian ukuran perangkat

2. Apa perbedaan font CDN dan next/font?

Font CDN mengambil font dari eksternal, sementara next/font menghost font secara lokal

3. Mengapa script bisa membuat website lambat?

Karena dengan adanya script website akan lambat dikarenakan harus membaca, mengunduh dan juga mengeksekusi kode javascript sebelum halaman sepenuhnya interaktif

4. Kapan harus menggunakan dynamic import?

Ketika ingin meuat komponen atau modul hanya pada saat dibutuhkan

5. Apa dampak bundle size terhadap UX?

Memperlambat waktu loading dan interaktifitas halaman

